

L2.5 Curve Sketching

§3.5 — precalc + the derivatives table

The two halves — run them in this order

① Precalc — *Buenos días*

D Domain

I Intercepts, and the *type* of each zero

A Asymptotes: VA, HA, SA — or **EB**

S Symmetry

② Calculus

f' ↗ ↘

CNs, and which are extrema

f'' ∩ ∪

CNs *and* PIPs, then the sign check

Neither half is new. The only thing §3.5 adds is running both on the same function and letting the sketch fall out.

ZERO TYPES — read them off the **FACTORED** form

B bounce

even power on the factor — touches the axis and turns back

PT pass through

odd power — crosses the axis

“keep BOTH forms on the page — factored for the zeroes, expanded for the derivatives”

EB — END BEHAVIOUR

EB is how the function behaves as it moves out toward $+\infty$ and toward $-\infty$. Polynomials have no asymptotes, so you might think the **A** row is empty. Put **EB** there instead — for a positive-leading odd-degree polynomial:

$$x \rightarrow -\infty \Rightarrow f(x) \rightarrow -\infty \quad x \rightarrow +\infty \Rightarrow f(x) \rightarrow +\infty$$

“the A row is never blank”

THE DERIVATIVES TABLE — one number line, five steps

① Find the **CNs** and the **PIPs**.

② Subdivide by **all of them together** — one line, not two.

③ Write the **evaluated** sign of f' and f'' on each piece.

④ **Draw it:** arrows above the line, cups below it.

⑤ **Sketch** f underneath, at the same horizontal positions.

$$\text{CN: } f'(c) = 0 \text{ or } f'(c) \text{ DNE} \quad \text{PIP: } f''(c) = 0 \text{ or } f''(c) \text{ DNE}$$

“a critical number is a shortlist, not an answer”

VERTICAL TANGENTS

Where f' **fails to exist** but the point is still in the domain, the curve has a **vertical tangent** — and it is drawn and labelled on the sketch like any extremum.

“a radical is where to expect one: $\sqrt[3]{x}(x-4)$ has one at the origin”

SLANT ASYMPTOTES

When long division leaves $f(x) = mx + b + R(x)$ and the remainder $R(x) \rightarrow 0$, the line $y = mx + b$ is a **slant asymptote**. Draw it dashed *on* the axes and sketch the curve against it.

“the whole argument is that the remainder dies”

WATCH OUT

~~X the A row is blank for a polynomial~~ → no VA and no HA is not “nothing” — EB goes there

~~X a graph can never cross an asymptote~~ → it cannot cross a VA; it *may* cross an HA or an SA

~~X $f''=0$ means inflection point~~ → it means PIP — *possible*. Confirm with a sign change

~~X a critical number is where $f'=0$~~ → or where f' DNE — that is the vertical tangent

~~X skip symmetry when it fails~~ → run the test and *report* it. “Neither” is an answer

~~X sketch from the expanded form~~ → the zeroes and their types come from the factored form

~~X $x=0$ goes on the number line twice~~ → once. One subdivision point, even when it is both a CN and a PIP